SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 10

Test Plan: Library Management System

AIM:

To identify and prepare positive test scenarios and test cases for the Railway Reservation System to verify that all major functionalities, such as train searching, seat availability checking, passenger detail entry, ticket booking, successful payment, PNR status checking, ticket downloading, and cancellation, work correctly with valid inputs and meet the specified functional requirements.

1. Introduction

This test plan defines the scope, approach, resources, schedule, and completion criteria for testing a Library Management System. The system manages books, members, issuing, returns, renewals, reservations, fines, and reports.

2. Objectives

- Verify that all required library functions work correctly.
- Check data accuracy for books, members, loans, and fines.
- Validate access control and protection of member information.
- Identify defects before the system is released.

3. Scope of Testing

In scope: login, member registration, book catalogue/search, book issue, return, renewal, reservation, fine calculation, inventory updates, reports, and logout.

Out of scope: third-party systems not integrated with the test environment, physical library equipment, and production data migration unless separately agreed.

4. Test Items and Requirements

Module	**Requirement to Verify**
Login and roles	Only authorized users access permitted features.
Book catalogue	Books can be added, edited, searched, and viewed.
Member management	Valid members can be registered and updated.
Issue and return	Loans update availability and due dates correctly.
Renewal and reservation	Renewals and reservations follow library rules.
Fines	Overdue fines follow configured policy.

Reports	Inventory, issue, return, and overdue reports are accurate.

5. Test Strategy and Methodology

Testing Type	**Purpose**
Functional	Verify features against requirements.
Integration	Check catalogue, member, loan, and fine modules together.
System	Test complete end-to-end workflows.
Regression	Re-run tests after changes and fixes.
Usability	Check screens and messages are understandable.
Performance	Check response times under expected load.
Security	Check authentication, authorization, sessions, and input validation.
Compatibility	Check supported browsers and screen sizes.

6. Test Design Techniques

- Equivalence partitioning: valid and invalid member IDs, book IDs, and inputs.
- Boundary value analysis: issue limits, fine limits, and due-date boundaries.
- Decision tables: issue eligibility based on member status, availability, and outstanding fines.
- State transition testing: available → issued → returned; reserved → available/issued.
- Use-case testing: register member, issue book, return book, and pay fine.

7. Test Environment and Test Data

Use a test server, test database, supported browsers, and test accounts with librarian/member roles. Use synthetic member and book records. Do not use real member passwords or sensitive production data.

8. Test Execution Strategy

- Review requirements and prepare traceability matrix.
- Design and review test cases; prepare test data.
- Run smoke tests to confirm the build is testable.
- Execute functional and integration tests, then system and non-functional tests.
- Record actual results and evidence for failures.
- Log defects, retest fixes, and run regression tests.
- Prepare a test summary report.

9. Entry and Exit Criteria

Entry Criteria	Exit / Completion Criteria
Approved requirements and test plan available.	All critical and high-priority cases executed.
Test environment and build available.	All planned tests have recorded results.
Test data and user accounts prepared.	No open critical defects; high defects resolved or formally accepted.
Smoke test passes.	Regression tests completed for fixes.
Team and tools ready.	Test summary reviewed and signed off.

10. Deliverables

- Test plan and test scenarios.
- Detailed test cases and test data.
- Execution logs and evidence.
- Defect reports and retest results.
- Traceability matrix and final test summary.

11. Roles and Responsibilities

Role	Responsibility
Test lead	Plan testing, assign work, review results, and report status.
Tester	Design and execute tests, record evidence, and report defects.
Developer	Investigate defects and deliver fixes.
Project owner	Clarify requirements and approve release risks.

12. Risks and Mitigation

Risk	Mitigation
Unclear library rules	Confirm issue limits, renewal rules, and fine policy.
Environment unavailable	Arrange access early and maintain a backup test window.
Incomplete test data	Prepare valid, invalid, and boundary data.
Late changes	Run impact analysis and regression testing.

13. Conclusion

This test plan provides a structured approach for evaluating the Library Management System. Following the strategy and completion criteria helps ensure that its functions are reliable, secure, and ready for acceptance.